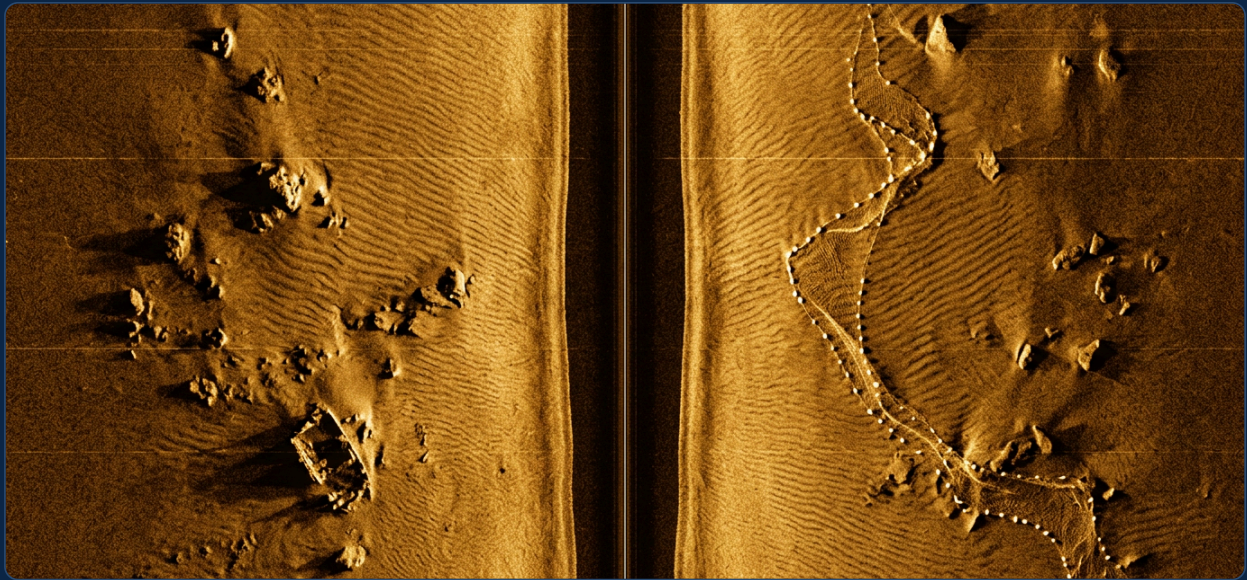


SagarNetra

A physics-grounded, offline AI system that finds ghost nets and marine debris in side-scan sonar, measures them in real metres, geotags them with an honest error radius, and proves where the survey could and could not see.



"Other systems draw boxes on sonar images. SagarNetra tells the dive team what the object is, how tall it is, where it is within a few metres, why it believes that, and which parts of the seabed are genuinely clear."

Document

Complete idea, solution design, uniqueness, dataset plan and build prompt

Inputs reviewed

Marine Debris Research Report and SonarTwin Solution Report

Prepared

26 September 2026

Contents

1	Executive summary	13	Training, evaluation and proof
2	The problem statement, decoded	14	Mapping to the graded deliverables
3	The eight real challenges	15	Dashboard and report design
4	Review of Report A: Marine Debris Research Report	16	Demo script for the judges
5	Review of Report B: SonarTwin Solution Report	17	Technology stack and hardware
6	Side-by-side verdict and what we carry forward	18	Build roadmap
7	The SagarNetra solution: core idea	19	Risks and mitigations
8	System architecture and flow charts	20	Impact, users and future scope
9	Module-by-module design	21	Final pitch
10	The physics and mathematics	A	Appendix: Antigravity build prompt (complete)
11	Uniqueness and value added	B	Appendix: References and data sources
12	Dataset plan: SagarNetra-SSS-IN v1		

How to read this document

Chapters 1 to 6 explain the problem and give an honest review of the two reports you received. Chapters 7 to 11 present the complete, improved solution and explain exactly what makes it unique. Chapters 12 to 19 turn the idea into a buildable plan: data, training, proof, demo, stack and risks. Appendix A is a copy-paste prompt for Google Antigravity that builds the entire prototype, including the dataset.

A note on honesty of numbers

Every performance number in this report is a **target**, not a measured result. The build prompt forces the prototype to measure and save its own metrics. Presenting targets as results in front of NIOT scientists is the fastest way to lose credibility, so the solution is designed to show measured evidence instead.

1 Executive summary

The problem. Abandoned, lost or discarded fishing gear ("ghost nets") keeps catching fish, turtles and dolphins for years, damages coral reefs, and fouls ship propellers. After cyclones, harbours and navigation channels also fill with new underwater obstructions: sunken boats, containers, collapsed structures. Side-scan sonar (SSS) can image the seabed in dark and murky water, but a human analyst must look through hours of noisy waterfall imagery. NIOT asks for an AI system that automates this: detect debris, score confidence, filter noise, geotag each object with latitude, longitude, size and class, and show it all on a live dashboard with downloadable reports, running offline on the survey vessel.

The hidden blocker. There is no public dataset of real ghost nets in side-scan sonar, and nothing public from Indian seabeds. A team that fine-tunes YOLO on crab-pot or shipwreck images will look good in its own demo and fail on real NIOT data. Whoever solves the missing-data problem, and can prove how reliable the system is, wins.

The two reports. Report A (Marine Debris Research Report) is an excellent research and risk memo: it correctly identifies the data gap, separates confidence into three kinds, and insists on "not enough evidence" as an output. But its unique value is mostly process, and it gives no way to fix the data gap in a hackathon. Report B (SonarTwin) fixes that with a physics simulator and adds physical outputs such as object height, but it remains high-level: it does not specify how simulated data is made realistic, how nets are specifically recognised, how confidence is calibrated, how coverage is proved, or how disaster mode works.

Our solution: SagarNetra. SagarNetra ("eye of the ocean") merges the strongest ideas of both reports and adds the missing engineering. It has seven pillars:

1. **SonarForge**, a physics simulator plus a **hybrid injector** that paints physically correct synthetic nets and debris (with matching shadows and speckle) into *real* seabed recordings. This creates the ghost-net training data that does not exist.
2. **A net-specific detector**: a sonar feature stack (intensity + shadow + line maps) feeding a small segmentation model, plus a dedicated **Net Signature Head** that finds the chain of bright floats and sinkers and the mesh texture that give a net away even when it casts no shadow.
3. **A Physics Verifier** that computes object height from shadow length and checks eight acoustic rules, rejecting rocks, interference, fish schools and surface echoes, and writes a human-readable reason for every decision.
4. **Calibrated Hazard Confidence** that fuses model probability, image quality, physics consistency, net evidence and number of views, and is checked with a reliability diagram so that "70 %" really means about 70 %.
5. **Survey-grade geotagging** with slant-range correction, towfish layback, heading and an explicit error budget, producing latitude, longitude and a 95 % error radius in metres, plus length, width and height.
6. **Coverage and Clearance mapping**: using the simulator, we measure the Probability of Detection (PoD) for every seabed cell, so the map shows not only what was found but which areas are genuinely clear and which need a second look.
7. **Disaster Mode**: before/after survey comparison that flags new obstructions after a cyclone, plus an anomaly head that catches "unknown man-made" objects no class list covers.

0.80

target ghost-net recall at 2 false
alarms per km²

≤ 5 m

target 95 % geolocation error with
standard GPS

≤ 0.05

target calibration error (ECE)

100 %

offline, CPU laptop or Jetson edge
device

Why this wins

It directly answers every item in the problem statement, it demonstrates capabilities judges will not see from other teams (height in metres, physics reasons, cleared-area map, before/after cyclone comparison), and every claim is backed by numbers the prototype measures itself.

2 The problem statement, decoded

Problem Statement 26057 comes from the National Institute of Ocean Technology (NIOT), Chennai, under the Ministry of Earth Sciences, in the Disaster Management theme. It asks for automated, AI-powered detection of underwater marine debris and anomalies using side-scan sonar imagery. The statement lists four expected components. The table below shows what each one says and what an expert evaluator will actually test.

Expected component	What the statement says	What evaluators will really check
1. Detection and segmentation model	Train a model (for example YOLO, Faster R-CNN or U-Net) to detect ghost nets, wreckage, pipes and other debris in SSS imagery.	Does it find faint, soft, draped nets next to rocks, sand ripples and seagrass? Wrecks and pipes are easy (bright, hard edges); nets are the hard case. Does it work on sonar it was not trained on?
2. Confidence scoring and noise filtering	Give a confidence score for each detection and filter out noise and false positives.	Is the score calibrated? Does the system deal separately with speckle, acoustic shadows, missing pings, interference and vessel motion? Does it explain why a detection was rejected?
3. Geotagging engine	Produce latitude, longitude, size and class of every detected object.	Does it read real sonar file headers (XTF/JSF), correct slant range, apply towfish layback and heading, and give size in metres with a stated position error?
4. Dashboard and reporting	Visualise detections with a live overlay on a map and allow report download.	Does it work offline, update as pings arrive, export JSON / CSV / GeoJSON / PDF, and produce something a dive or ROV team can use?

The implicit requirements

- **Edge and offline operation.** Survey vessels often have no internet. The system must run on a laptop CPU or a small GPU board such as an NVIDIA Jetson.
- **Disaster management.** The theme signals post-cyclone and post-tsunami surveys of ports and channels (Chennai, Visakhapatnam, Paradip, Kochi). The system must flag *new* and *unknown* man-made objects, not only nets.
- **Operational trust.** A false "all clear" in a shipping channel is dangerous. The output must say where the survey was able to see and where it was not.
- **Real data formats.** NIOT and survey contractors record in XTF (eXtended Triton Format) or vendor formats such as EdgeTech JSF. Screenshots alone are not enough.

What side-scan sonar images actually look like

A towfish sends acoustic pulses sideways to port and starboard. Each ping produces one line of echo strength versus time (slant range); stacking pings as the vessel moves forms a "waterfall" image. Hard or rough surfaces facing the sonar appear bright; areas hidden behind raised objects receive no sound and appear as dark **acoustic shadows**. The shadow length encodes object height, which is the single most valuable physical clue in SSS. Images also contain a dark water-column band at the centre (nadir), speckle noise, gain changes with range, and geometric distortion from slant range and vessel motion.

Key insight

Side-scan images are produced by simple, well-understood acoustic geometry. That is a weakness for pure computer-vision approaches (the images do not look like photos) but a huge opportunity: the physics can be simulated, inverted to measure height, and used to verify detections.

3 The eight real challenges

A good solution must be designed around the problems that actually make this task hard. Each challenge below is mapped to the SagarNetra component that addresses it.

#	Challenge	Why it is hard	SagarNetra answer
1	No ghost-net SSS data	Public SSS datasets contain crab pots, traps, ships and aircraft. None contain real gillnets on Indian seabeds, and most have no navigation metadata.	SonarForge simulator + hybrid injection into real seabed recordings (Ch. 9.3).
2	Domain gap between sonars	Different frequencies (100 kHz to 1 MHz), ranges, gains and vendors change pixel size and texture completely.	Slant-range and along-track resampling to a fixed 0.10 m ground grid, gain normalisation, heavy physics-based domain randomisation.
3	Nets are faint	Nylon twine is a weak acoustic reflector; flat nets cast almost no shadow. Only floats, sinkers, head/foot ropes and mesh texture are visible.	Net Signature Head: float-chain finder, rope ridge detector, mesh texture energy (Ch. 9.5).
4	Look-alike confusers	Rock clusters, sand ripples, seagrass, fish schools and biological mounds create bright/dark patterns similar to debris.	Confuser classes trained explicitly + Physics Verifier rules (Ch. 9.6).
5	Noise and artefacts	Speckle, missing pings, electrical interference, port/starboard crosstalk, surface multipath, heave and yaw.	Ping quality control, despeckling, artefact rules, image-quality score in confidence.
6	Accurate position and size	Pixels are in slant range; the towfish trails behind the GPS antenna; heading errors grow with range.	Full geometric chain with layback and error budget, reporting a 95 % radius (Ch. 9.8).
7	"Nothing found" is not "nothing there"	A net may be missed at long range, near nadir, on rough seabed or in a dropout. Reporting only detections implies false clearance.	Probability-of-Detection coverage map and second-look planner (Ch. 9.9).
8	Edge deployment	Real-time processing must keep up with ping rate on modest hardware without internet.	Cheap CFAR pre-screen, small ONNX model, tiles only where needed; CPU-first design.

The trap most teams will fall into

Train YOLO on a public SSS dataset, draw boxes on test images from the same dataset, place pins on a map with made-up coordinates, and call it done. It demos well and says nothing about ghost nets, calibration, geolocation accuracy or real surveys. SagarNetra is designed to be the opposite of this.

4 Review of Report A: Marine Debris Research Report

What it is. A research and product-opportunity report covering the environmental problem, why cameras fail underwater, why manual interpretation is slow, existing precedents (GhostNetZero, GhostVision-type systems, commercial tools such as SonarWiz and SeeByte), Indian stakeholders, a detailed dataset survey with licences, and a set of "differentiators" centred on verification and honesty.

Strengths we keep

Strength	Why it matters
Honest market view	It admits that "detect and put a pin on a map" already exists commercially. This forces a real differentiator.
Correct diagnosis of the data gap	It shows that no official ghost-net SSS dataset exists and that public sets differ in sensor, target and geography.
Three kinds of confidence	Separating class confidence, observation quality and position accuracy is the right structure. We keep it and make it quantitative.
"Insufficient evidence" as an output	Reporting what the survey could not see is operationally important. We turn it into a measured PoD map.
Object-shadow-seabed reasoning	It points to shadow geometry. We implement it as a working Physics Verifier with eight rules.
Target relocation and field feedback	Useful ideas; we implement them as a PDF work order with a search box and a review loop for active learning.
Dataset list and licence warnings	Accurate and valuable. We adopt the list and add a concrete role for each dataset.

Gaps that made it unsatisfying

1. **Its unique value is process, not technology.** "Verify first", "recommend a second look", "learn from field results" are workflow rules that any team can copy in an afternoon.
2. **It names the fatal problem but does not solve it.** Its remedy for the missing data is "partner surveys" and "ROV verification", which a hackathon team does not have.
3. **The physics is described, never built.** Shadow geometry is mentioned but never becomes a visible output such as object height.
4. **Second look cannot be demonstrated** without a vessel, unless its benefit is estimated in simulation.
5. **Weak mapping to the four graded deliverables;** following it produces a strategy paper rather than a scored prototype.
6. **No build specification:** no model architecture, feature design, training recipe, evaluation protocol or demo plan.

Verdict

An excellent risk memo and research foundation (about 8/10 for research) but a weak product specification (about 4/10 for buildability and uniqueness).

5 Review of Report B: SonarTwin Solution Report

What it is. A shorter solution report that reviews Report A, then proposes "SonarTwin": a physics-based side-scan simulator that manufactures ghost-net training images, a two-stage detector (classical CFAR plus a small neural network and an anomaly head), a physics module that estimates height from shadows, three confidence scores, a geotagging engine and a dashboard.

Strengths we keep

- **It attacks the real blocker.** Simulation is the only practical way to get labelled ghost-net sonar data inside a hackathon.
- **Physical units instead of pixels.** Height from shadow length, footprint in metres, position error in metres: outputs that bounding-box teams cannot give.
- **The simulator as a test bench.** Measuring the operating envelope (small nets, rough seabeds, long range, low signal) in software is a strong idea.
- **Two-stage detection with an anomaly head** that catches unknown man-made objects after disasters.
- **Good mapping to deliverables** and a sensible demo mindset.

Gaps we fix

Gap in SonarTwin	How SagarNetra fixes it
Pure simulation creates a sim-to-real gap	Hybrid injection: synthetic objects are rendered into real seabed recordings with geometry-consistent shadows and matched speckle statistics, so the model learns on real textures.
Nets treated like any other object	A dedicated Net Signature Head for float/sinker chains, ropes and mesh texture, because draped nets often have no shadow.
Physics module only estimates height	A full Physics Verifier with eight rules (order, mirror interference, water column, nadir, surface multipath, persistence, resolution, slope) and a written reason log.
Three separate scores, no calibration	One calibrated Hazard Confidence with transparent components, fitted on validation data and checked with ECE and a reliability diagram.
Operating envelope as a chart only	PoD look-up table used live to colour every seabed cell as clear, insufficient or not surveyed.
Disaster theme only mentioned	Full Disaster Mode: before/after registration and change detection with three demo scenarios at Indian ports.
No Indian context in data	Seabed presets modelled on Indian coastal settings and demo surveys placed at six Indian sites.
No XTF demo data	A minimal XTF writer so simulated surveys become real XTF files with navigation, exercising the real ingestion path.
No build specification	Complete module design (Ch. 9), dataset recipe (Ch. 12), evaluation protocol (Ch. 13) and a full AntigraVity prompt (Appendix A).

Verdict

The right core idea (about 7/10) but under-specified. Its simulator alone would not transfer reliably to real data, and its uniqueness claims need concrete, demonstrable features.

6 Side-by-side verdict and what we carry forward

Criterion	Report A (Research)	Report B (SonarTwin)	SagarNetra (this report)
Problem understanding	Very strong	Strong	Strong, with eight explicit challenges
Solves the data gap	No	Yes, simulation only	Yes: simulation + hybrid injection + real public data + negatives
Ghost-net specific detection	No	Partially	Yes: Net Signature Head
Noise and false-positive filtering	Described	Two-stage filter	Quality control + despeckle + confusers + 8-rule verifier
Confidence	Three concepts	Three scores	One calibrated score with visible components
Geotagging	Principles	Outline	Full chain with layback, error budget and 95 % radius
Size in metres (incl. height)	No	Yes	Yes, plus orientation, area and outline polygon
Coverage / clearance	Concept	Envelope chart	Live PoD map with km² clear / insufficient
Disaster management	Mentioned	Mentioned	Before/after change detection + unknown-object head
Buildability in a hackathon	Low	Medium	High: milestone plan and build prompt

Design principles for the final solution

1. **Physics first, deep learning second.** Use acoustic geometry to create data, verify detections and measure objects; use neural networks where pattern recognition is genuinely needed.
2. **Train on reality wherever possible.** Real seabed textures with synthetic targets beat purely synthetic images.
3. **Nets are the priority class.** Everything else (pipes, wrecks, cylinders) is easier; the design must be judged on nets.
4. **Every output is explainable.** Each detection carries its evidence and rule log.
5. **Absence must be measured.** "Clear" is only reported where PoD is high.
6. **Measure, do not claim.** All metrics are produced by the prototype and saved.
7. **Offline and edge-first.** CPU laptop by default; Jetson as an upgrade.

7 The SagarNetra solution: core idea

In one sentence

SagarNetra manufactures the ghost-net sonar data that does not exist by painting physically correct synthetic nets into real seabed recordings, detects them with a net-aware model, verifies each detection with acoustic physics, and reports calibrated confidence, true position and size in metres, and a measured map of what the survey could and could not see.

The seven pillars

1. SonarForge + Hybrid Injection

Physics simulator for seabeds, nets, ropes, pipes, cylinders, wrecks, traps and confusers; injector that renders them into real SSS backgrounds with matching shadows and speckle.

2. Net-aware Detection

Three-channel sonar feature stack (intensity, shadow probability, line map) into YOLO11n-seg, plus the Net Signature Head and a PatchCore anomaly head.

3. Physics Verifier

Height from shadow; eight rules that accept, reject or flag detections, each with a written reason.

4. Calibrated Hazard Confidence

One 0-100 % score fusing calibrated probability, observation quality, physics, net evidence and views; validated with a reliability diagram.

5. Survey-grade Geotagging

Slant-range correction, layback, heading, UTM conversion, 95 % error radius, length, width, height, orientation and outline.

6. Coverage and Clearance (PoD)

Simulation-measured Probability of Detection per 1 m cell; clear / insufficient / not surveyed; second-look line planner.

7. Disaster Mode

Before/after survey registration and change detection, producing a prioritised list of new obstructions after cyclones.

Delivered through

An offline dashboard with live waterfall and map, a target inspector, and exports: JSON, CSV, GeoJSON, KML and a PDF work order for dive/ROV teams.

Why this is logical, step by step

1. The biggest blocker is the absence of labelled ghost-net sonar data.
2. Side-scan images come from simple, known physics, so they can be simulated accurately.
3. Pure simulation still looks different from real data, so we inject simulated objects into real seabed recordings: real texture, synthetic target, perfect label.
4. Because we control the physics, every training object also has its true height, size and position, so we can train and test physical measurements, not just boxes.
5. The same physics runs backwards at inference time: shadow length gives height, geometry gives position, and rules expose false alarms.
6. The simulator also measures how often a net of a given size is found at a given range and seabed roughness. That table turns an empty map into a statement of clearance.
7. Comparing surveys before and after a cyclone uses the same pipeline, delivering disaster-management value directly.

8 System architecture and flow charts

Flow chart 1: live survey pipeline (on the vessel)

INPUT

Sonar logs

XTF / JSF / SNL files or live ping stream

Image logs

PNG / TIFF / GeoTIFF + navigation CSV

Navigation

GPS, heading, cable out, attitude, altitude

PREPROCESSING AND NOISE FILTERING

Ping QC

Dropouts, altitude jumps, nav gaps

Bottom track

First return + Kalman smoothing

Geometry

Slant-range and along-track to 0.10 m grid

Gain + despeckle

Normalisation, enhanced Lee filter

Feature stack

Intensity, shadow, line map; 512 px tiles

DETECTION

Stage A: CFAR

Highlight-shadow pairs, fast screen

Stage B: YOLO11n-seg

6 hazard + 4 confuser classes

Net Signature Head

Float chains, ropes, mesh texture

Anomaly head

PatchCore: unknown man-made

VERIFICATION AND SCORING

Physics Verifier

Height from shadow + 8 rules + reasons

Hazard Confidence

Calibrated fusion of 5 evidence terms

Status

Candidate / Low evidence / Rejected

GEOLOCATION AND COVERAGE

Geotag

Layback, heading, UTM, 95 % radius

Dimensions

Length, width, height, orientation

Multi-view association

Merge across survey lines

PoD coverage map

Clear / insufficient / not surveyed

OUTPUT

Live dashboard

Waterfall + map + inspector

Reports

JSON, CSV, GeoJSON, KML, PDF work order

Second look

Suggested lines, GPX export

Disaster Mode

Before/after new obstructions

Flow chart 2: data and learning loop (on shore)

SonarForge simulator

Indian seabed presets, nets, debris, confusers, noise, motion

Hybrid injector

Synthetic targets into real seabed tiles

Real public data

Ghost Pot, SCTD, KLSG-II, AI4Shipwrecks, NOAA/USGS

SagarNetra-SSS-IN v1

Sim + hybrid + real + negatives; leak-free splits

Curriculum training

Sim + hybrid, then fine-tune with real

Calibration

Temperature scaling + fusion weights

PoD sweep

Full pipeline on reference targets

ONNX / TensorRT export

Laptop CPU or Jetson

Field review

Confirm / reject in the inspector

Flywheel

Confirmed targets re-simulated as thousands of variations

9 Module-by-module design

9.1 Ingestion

Readers for **XTF** (via pyxrf: sonar channels, navigation, attitude, altitude), **EdgeTech JSF** (sonar data messages), our own **SNL** HDF5 format, and **image logs** (PNG / TIFF / JPG waterfalls with a sidecar navigation CSV, or GeoTIFF mosaics). When no navigation exists, the system runs in **pixel-only mode**: detections are shown in the image, position and dimensions are left empty with a stated reason, and coordinates are never invented. A replay source streams any file at 1x to 50x speed to simulate a live survey.

9.2 Preprocessing and noise filtering

Step	What it does and why
Ping quality control	Flags dropouts (zero, constant or very low energy pings), altitude jumps above 0.5 m, navigation gaps above 2 s and excessive attitude rates. Single dropouts are interpolated; bursts are marked NO_DATA so the model never hallucinates content.
Bottom tracking	Detects the first seabed return on each side from the smoothed energy gradient and smooths altitude with a Kalman filter; disagreements with the header altitude are reported.
Slant-range correction	Converts slant range to ground range and removes the water column, then resamples to a fixed 0.10 m grid. This makes different sonars and ranges look alike to the model.
Motion compensation	Resamples pings by distance travelled, compensates heading and yaw; per-ping altitude handles heave.
Gain normalisation	Removes range-dependent brightness using a sliding mean per ground-range bin, then log-compresses.
Despeckling	Enhanced Lee or SRAD filter that reduces speckle while keeping edges; the raw layer is retained for evidence.
Feature stack	Instead of fake RGB, three meaningful channels: (0) normalised intensity, (1) CFAR shadow probability, (2) ridge/line map. This hands the network the two cues that matter most in sonar: shadows and thin lines.

9.3 SonarForge simulator and hybrid injector

Seabed. A 2.5D height and material map at 5 cm resolution combining slopes, sand ripples, rock clusters, seagrass and mud. Five presets represent Indian settings as parameter ranges: East-coast sand shelf, West-coast mud, reef rubble, seagrass bed and harbour floor.

Objects. Ghost nets are modelled as real gear: headrope and footrope lines with a mesh panel, floats every 0.5 to 2 m, sinkers, weak nylon backscatter with mesh-period texture, draping over the seabed, 0 to 70 % burial, bundled masses, snags on rocks and tears. Also ropes, pipes, cylinders/drums, wreck debris (boats, containers, structures), traps, and confusers (rocks, ripples, seagrass, fish schools, mounds) that are trained as negatives.

Rendering. Ray casting across the seabed profile for each ping using the sonar equation: spreading and absorption loss, Lambertian backscatter by material, occlusion shadows, time-varied gain, K-distributed speckle, noise floor, nadir gap, surface multipath, interference stripes, crosstalk ghosts and ping dropouts. Heave, pitch, roll, yaw and speed variations are applied before rendering, and lawn-mower survey lines are laid out around six Indian demo sites.

Hybrid injection. Real seabed-only tiles are taken from public datasets. The injector estimates the local geometry (nadir, altitude, range direction), fits local backscatter and speckle statistics, renders the synthetic object's highlight and a shadow with the *same* geometry, darkens the shadow to the local noise floor, re-samples speckle to match and blends the edges. The result: real seabed texture with a physically correct synthetic target and a perfect label.

Why hybrid injection is the decisive step

Models trained on pure simulation learn simulator artefacts. Models trained only on public real data never see a ghost net. Hybrid tiles give both: real texture statistics and the target we need, plus its ground-truth height, size and position.

9.4 Detection (Stage A, Stage B and anomaly)

- **Stage A, CFAR screen:** order-statistic CFAR finds bright highlights; inverse CFAR finds shadows; highlights paired with a shadow on the far-range side become candidates. It is cheap and runs on every ping.
- **Stage B, YOLO11n-seg:** instance segmentation on the feature stack, 512 px tiles, exported to ONNX. Hazard classes: ghost_net, rope_cable, pipe, cylinder, wreck_debris, trap_pot. Confuser classes (trained, never reported as hazards): rock_cluster, sand_ripple, seagrass, fish_school.
- **Anomaly head:** PatchCore-lite on frozen ResNet18 features from seabed-only tiles. High anomaly with no hazard or confuser class becomes *unknown_manmade*, which is exactly what post-disaster surveys need.
- **Candidate fusion:** union of all sources with mask-based non-maximum suppression.

9.5 Net Signature Head (ghost-net specialist)

A flat gillnet on the seabed often has no shadow and a faint body, but it has a distinctive structure. The head measures four pieces of evidence and combines them into a net score S_{net} between 0 and 1:

1. **Point targets:** small bright blobs of 8 to 50 cm (floats and sinkers) found with a Laplacian-of-Gaussian detector.
2. **Chains:** the points are connected into a graph, a minimum spanning tree is built and the longest smooth paths are scored for regular spacing (coefficient of variation below 0.35) and smooth curvature.
3. **Lines:** ridge skeletons of at least 2 m (headrope, footrope, ropes).
4. **Mesh texture:** Gabor filter energy tuned to the mesh period.

The head also produces a human-readable explanation, for example: *"chain of 14 bright points, mean spacing 0.9 m, regular; 11 m rope ridge; mesh texture present"*. Rocks produce irregular clusters, ripples produce straight parallel lines, and neither produces a regular float chain, which is why this head sharply reduces false alarms on the priority class.

9.6 Physics Verifier

For each detection the verifier measures highlight extent, shadow length L_s , ground range x_o and altitude H , computes the object height and applies eight rules. Each rule returns consistent (+1), not assessable (0) or inconsistent (-1) with a reason, and the average forms V_{phys} .

Rule	Check	Purpose
R1	Class height prior	Net 0-1.5 m, rope 0-0.3 m, pipe 0.1-1.5 m, cylinder 0.2-2 m, wreck 0.5-15 m, trap 0.3-1.2 m. Class-aware: flat nets without shadow are not penalised.
R2	Highlight before shadow	A real seabed object has its shadow on the far-range side. Reversed order indicates an artefact or depression.
R3	Port/starboard mirror	The same return at the same ping and range on both sides is crosstalk or interference.
R4	Along-track persistence	A true object persists over the pings predicted by beam footprint; single-ping spikes are noise.
R5	Water column	Returns before the first seabed echo are fish schools or suspended material, not seabed debris.
R6	Nadir and surface bands	Returns in the nadir band or at about twice the depth are multipath artefacts.
R7	Resolution adequacy	The object must span at least three beam footprints; otherwise the evidence is insufficient.
R8	Local slope	Slopes above 5 degrees make height unreliable and widen the position error.

9.7 Calibrated Hazard Confidence

The class probability is first calibrated with temperature scaling. Then five evidence terms are fused with weights learned by logistic regression on validation data (Ch. 10). The dashboard shows the final score and each component, so an operator sees *why* a target scored 82 %. Status: **CANDIDATE** at 60 % or more, **LOW EVIDENCE** at 35 to 60 %, **REJECTED** below 35 % (kept in the log), and **INSUFFICIENT** for coverage cells that could not be assessed.

9.8 Geotagging and dimensions

The engine converts each detection from pixels to the Earth. It projects GPS positions to UTM (zones 43, 44 or 45 for Indian waters), computes the towfish position from layback (given, or from cable out and depth), offsets the object across-track by its ground range to port or starboard, interpolates along-track by timestamp, and converts back to WGS84 latitude and longitude. A full error budget (GPS, layback, heading, range and timing) gives a **95 % error radius r95** in metres. Dimensions come from the oriented minimum rectangle of the ground-range mask: length, width, orientation and area, with height from the verifier. Detections from different survey lines are merged when their error ellipses overlap, and the number of independent views is recorded.

9.9 Coverage, Probability of Detection and second look

Using SonarForge, the complete detection pipeline is run thousands of times on two reference targets (a 5 m x 2 m gillnet fragment 50 % buried, and a 0.6 m x 1.5 m cylinder) across ground range, altitude ratio, seabed roughness and signal-to-noise ratio. The resulting PoD table is used live: every 1 m seabed cell receives the best PoD across passes and is coloured **surveyed clear** (PoD at least 0.8 and no detection), **insufficient** (PoD below 0.5 or quality flags), **not surveyed**, or **candidate**. The summary reports km² in each state. For low-evidence targets and insufficient areas, the planner proposes a new survey line with a perpendicular heading that places the target at 30 to 60 % of range, exported as GeoJSON and GPX.

9.10 Disaster Mode

Before and after surveys are georeferenced and finely aligned (ECC image registration on the overlap). A change map combines the normalised log-ratio of intensities with anomaly-feature distance. Changed regions larger than 1 m² are labelled **NEW_OBSTRUCTION** or **REMOVED** and pass through the same verifier, geotagging and reporting. Three synthetic scenarios are provided: Chennai Port basin (sunken fishing boat, container, net mass), Kochi channel (collapsed pier blocks, displaced pipe) and Visakhapatnam fishing harbour (capsized boats and ropes).

9.11 Reports and review

Each detection is exported with ID, class, status, Hazard Confidence and its components, rule log, position with r95 and method (or empty with reason), dimensions, pixel bounding box, WGS84 outline, views, time, source file, priority, recommended action and review status. Formats: JSON, CSV, GeoJSON (points, outlines, coverage, second-look lines), KML and a **PDF work order** with a ranked target list and one evidence page per target including a search box of side 2 x r95 for divers or an ROV. Operator reviews (confirm, reject, change class, unsure) are stored for active learning.

$$\text{Priority} = (C / 100) \times \text{severity}(\text{class}) \times 1 / (1 + r95 / 10)$$

Severity is multiplied by 1.5 when the object lies within 200 m of a shipping channel or reef polygon supplied by the user.

10 The physics and mathematics

Slant range to ground range

$$x = \sqrt{r^2 - H^2}$$

r = slant range from the two-way travel time ($r = c \cdot t / 2$, $c \approx 1500$ m/s), H = towfish altitude, x = horizontal ground range.

Object height from shadow length

$$h = H \cdot L_s / (x_o + L_s)$$

L_s = shadow length on the ground, x_o = ground range to the object's far edge. Derived from similar triangles between the sonar, object top and shadow end.

Rendered echo intensity (simulator)

$$I_{dB} = SL - 2 \cdot TL(r) + BS(\theta, \text{material}) + B(\theta_{dep}), \quad TL(r) = 20 \cdot \log_{10}(r) + \alpha \cdot r, \quad BS = \mu + 10 \cdot \log_{10}(\cos^2 \theta)$$

$\alpha \approx 0.10$ dB/m at 450 kHz and 0.30 dB/m at 900 kHz; μ is the Lambert coefficient of the material (e.g. sand -27 dB, rock -17 dB, metal -5 dB, nylon twine -30 dB, air float -3 dB).

Target position

$$E = E_f + s \cdot x \cdot \cos \psi, \quad N = N_f - s \cdot x \cdot \sin \psi$$

(E_f , N_f) = towfish UTM position after layback, ψ = heading clockwise from north, $s = +1$ starboard and -1 port.

Position error budget

$$\sigma^2 = \sigma_{gps}^2 + \sigma_L^2 + (x \cdot \sigma_\psi)^2 + \sigma_x^2 + (v \cdot \sigma_t)^2, \quad r95 = 2.45 \cdot \sigma$$

Defaults: $\sigma_{gps} = 2$ m, $\sigma_L = 0.1 \cdot L + 1$ m, $\sigma_\psi = 1^\circ$, $\sigma_t = 0.1$ s. The factor 2.45 converts a 2D Gaussian σ to a 95 % circular radius.

Hazard Confidence

$$C = 100 \cdot \text{sigmoid}(w_0 + w_1 \cdot \text{logit}(p_{cal}) + w_2 \cdot Q_{obs} + w_3 \cdot V_{phys} + w_4 \cdot S_{net} + w_5 \cdot \min(\text{views}, 3))$$

p_{cal} = temperature-scaled class probability; Q_{obs} = observation quality (dropouts, altitude ratio, range band, attitude, resolution); V_{phys} = verifier score; S_{net} = net evidence. Weights fitted on validation data and checked with Expected Calibration Error (ECE).

Along-track resolution check

$$\text{footprint} \approx x \cdot \theta_h, \quad \text{require object length / footprint} \geq 3$$

θ_h = horizontal beamwidth in radians (about 0.4° at 450 kHz, 0.2° at 900 kHz). At 50 m range and 0.4° , the footprint is about 0.35 m.

11 Uniqueness and value added

Uniqueness must be something judges can see in the demo and that another team cannot add in an afternoon. The table lists ten differentiators, what the judge sees, and why it is hard to copy.

#	Differentiator	What the judge sees	Why it is hard to copy
1	Hybrid injection data engine	Real seabed tiles with synthetic ghost nets whose shadows and speckle match the background.	Needs acoustic geometry, speckle statistics and a simulator working together.
2	Net Signature Head	A half-buried net found through its float chain, with a written explanation.	Requires domain knowledge of gillnet construction, not generic detection.
3	Height in metres from shadow	A height diagram for each target: "wreck 2.1 m tall".	Needs correct slant-range geometry and bottom tracking.
4	Physics-explained rejections	A rock or interference echo rejected with a reason: "port/starboard mirror at ping 4312".	Eight rules tied to acoustic behaviour, not thresholds on scores.
5	Calibrated confidence	A reliability diagram before and after calibration; components shown per target.	Needs a proper validation split and fusion fitting.
6	Honest 95 % position radius	Each pin has an error circle; the PDF has a search box for divers.	Requires layback, heading and a full error budget.
7	Measured clearance map	Green = clear, amber = insufficient, grey = not surveyed, with km ² totals.	Only possible with a simulator that measures PoD for the full pipeline.
8	Disaster Mode	Before/after cyclone swipe with a list of new obstructions at Indian ports.	Needs registration, change detection and the unknown-object head.
9	Real XTF end to end	Genuine XTF files with navigation stream into the dashboard.	Most teams only handle PNG screenshots.
10	Interactive SonarForge Lab	Sliders for burial, range and roughness; the PoD curve updates live.	Makes the operating envelope tangible and educational.

Positioning against existing approaches

Capability	Typical hackathon YOLO	Commercial SSS software	SagarNetra
Ghost-net training data	None (crab pots)	Proprietary / manual	Hybrid physics data
Ghost-net specific logic	No	Limited	Net Signature Head
Height and size in metres	No	Manual measuring tools	Automatic
Explained false-alarm rejection	No	No	8-rule verifier
Calibrated confidence	Raw score	Varies	Calibrated, with components
Measured clearance	No	Coverage only	PoD per cell
Before/after disaster change	No	Manual comparison	Automatic
Offline edge operation	Sometimes	Yes, licensed	Yes, open stack

Value to each stakeholder

- **NIOT / MoES:** faster survey interpretation, a reusable simulator and dataset for Indian waters, alignment with the Deep Ocean Mission.
- **Port authorities (Chennai, Kamarajar, Visakhapatnam, Paradip, Kochi, JNPT):** rapid post-cyclone channel clearance with measured confidence.
- **Coast Guard and Navy hydrographers:** prioritised obstruction lists with search boxes.
- **Fisheries departments and fishing communities:** gear-loss hotspot maps and retrieval programmes that protect fish stocks and livelihoods.
- **Marine conservation (Gulf of Mannar, Lakshadweep, Andaman):** reef protection from entangling nets.

12 Dataset plan: SagarNetra-SSS-IN v1

12.1 Public datasets and their roles

Dataset	Content	Role in SagarNetra	Licence / access
Ghost Pot Side-Scan Sonar Detection Dataset	About 6,674 Humminbird SSS images with crab-pot bounding boxes (derelict fishing gear)	Real trap_pot class; negative seabed regions for injection	Hugging Face, gated; check licence (GPL noted)
SCTD 1.0	Real SSS images of ships, aircraft and human bodies, Pascal VOC boxes	wreck_debris (ship / aircraft) real examples	GitHub, research use
SCTD2	Klein SSS images of small traps, segmentation masks	trap_pot masks; boxes derived from masks	GitHub, research use
SeabedObjects-KLSG-II	Ship, airplane and seafloor SSS images (seafloor class about 578)	Wreck examples; seafloor backgrounds for injection and negatives	GitHub, research use
AI4Shipwrecks	286 SSS images from 28 wreck sites with pixel masks	Real wreck segmentation; held-out real test	University of Michigan release
S3Simulator	About 7,721 synthetic SSS ship and plane images	Pre-training and simulator comparison only	Public research release
NOAA NCEI / USGS side-scan archives	Raw survey data (often XTF) with navigation	Backgrounds, ingestion and geotag tests	US public domain; cite source
SeafloorAI (optional)	Large seabed mapping dataset	Texture statistics only	CC BY-NC-SA 4.0, request-based
Marine Debris FLS (optional)	Forward-looking sonar debris images	Kept separate; not mixed into SSS training	Zenodo

12.2 Composition of SagarNetra-SSS-IN v1

Part	Size	Description
S	20,000 tiles	Fully simulated tiles across all five seabed presets, all classes, 450 / 900 kHz and 30 / 50 / 75 m ranges, with physical ground truth.
H	15,000 tiles	Hybrid: synthetic objects injected into real seabed backgrounds.
R	All available	Real labelled tiles re-mapped: crab pots and traps to trap_pot; ships and aircraft to wreck_debris with subtype.
N	10,000 tiles	Real seabed-only negatives for false-alarm suppression and the anomaly bank.
D	300 pairs	Synthetic before/after disaster pairs.
X	6 logs	Synthetic XTF surveys with navigation at Chennai, Kochi, Visakhapatnam, Gulf of Mannar, JNPT Mumbai and Paradip.

Labels: YOLO segmentation text files, COCO JSON and per-tile metadata (source, preset, sensor, frequency, resolution, altitude, SNR, roughness and physical truth for every object). **Splits** are made by scene seed, survey, site and physical object, never by random crops of the same image, and an entire real held-out set (unseen sites) is kept untouched until final evaluation.

12.3 Physically valid augmentation

Allowed

Port/starboard mirror across nadir (shadows still point away), along-track flip, along-track stretch 0.8-1.2, gain and TVG jitter, gamma, speckle re-sampling, dropout stripes, range-dependent blur, rotation within 5 degrees.

Forbidden (breaks sonar physics)

90 or 180 degree rotations, flips that make shadows point toward nadir, colour jitter, mosaics that mix sides or ranges.

13 Training, evaluation and proof

Training recipe

- **Curriculum:** 60 epochs on simulated + hybrid tiles, then 40 epochs of fine-tuning including real data (oversampled three times).
- **Model:** YOLO11n-seg, image size 512, AdamW, cosine learning rate from 1e-3, early stopping on mask mAP50-95.
- **Quick mode:** 3,000 tiles and a few epochs on CPU, so the prototype is always demonstrable.
- **Anomaly bank** from seabed-only tiles; **calibration** of temperature and fusion weights on a dedicated split; **export** to ONNX with numerical equivalence checks, optional INT8 TensorRT on Jetson.

Metrics and targets

Metric	Target	How it is measured
Ghost-net recall	≥ 0.80 at ≤ 2 false alarms / km ²	Hybrid test split, per km ² of surveyed area
Mask mAP50 / mAP50-95	Reported per class and source	Separate sim, hybrid and real test sets
Expected Calibration Error	≤ 0.05	15-bin reliability diagram, before and after
Geolocation CEP95	≤ 5 m with standard GPS	Against synthetic ground truth
Height error	Mean absolute error reported	Objects taller than 0.3 m
Latency	≤ 150 ms per 512 px tile on laptop CPU	Timed in pipeline; pings per second shown live
PoD curves	Published	Versus range, burial and roughness

Ablation study (the strongest proof slide)

Each component is added one at a time and measured: (1) YOLO on single-channel intensity, (2) + sonar feature stack, (3) + Net Signature Head, (4) + Physics Verifier, (5) + anomaly head, (6) + confidence fusion. The ablation shows judges exactly how much each idea contributes, rather than asking them to trust the design.

Avoiding misleading validation

No test tile shares a scene, survey or physical object with training. Real held-out sites are evaluated once at the end. Simulated-only results are always labelled as such.

14 Mapping to the graded deliverables

Deliverable	SagarNetra implementation
1. Detection / segmentation model	Two-stage CFAR + YOLO11n-seg on a sonar feature stack, Net Signature Head for ghost nets, anomaly head for unknown man-made objects; trained on SagarNetra-SSS-IN v1.
2. Confidence scoring and noise filtering	Ping quality control, gain normalisation, despeckling, confuser classes, eight-rule Physics Verifier and calibrated Hazard Confidence with visible components and status states.
3. Geotagging engine	XTF/JSF ingestion, slant-range and layback correction, UTM projection, latitude/longitude with 95 % error radius, length, width, height, orientation, class, multi-view merging.
4. Dashboard and reports	Offline React dashboard with live waterfall, map overlay, coverage layer, inspector, disaster comparison and SonarForge Lab; JSON, CSV, GeoJSON, KML and PDF work-order downloads.
Theme: Disaster Management	Disaster Mode before/after comparison, new-obstruction list and prioritised clearance for ports.
Edge deployment	ONNX Runtime on CPU; optional TensorRT on Jetson Orin Nano; no internet at runtime.

15 Dashboard and report design

Screen	Content
Mission Control	Upload; dual port/starboard waterfall scrolling live with coloured masks; map with ship and towfish tracks, detection pins with error circles and coverage layer; counters for pings, km ² , detections per class, pings per second and device.
Target Inspector	Crop with mask and shadow overlay, height diagram, confidence component bars, rule log and review buttons (confirm, reject, change class, unsure).
Coverage and Clearance	PoD heat map for the net and cylinder references, second-look lines, clearance summary in km ² .
Disaster Compare	Before/after swipe slider and NEW_OBSTRUCTION list with zoom-to.
SonarForge Lab	Sliders for seabed preset, net length, burial, range, altitude and SNR with a live generated waterfall and PoD curve.
Reports	Downloads in all formats with a JSON preview.

Visual style: a dark operations console (navy background, teal accent, amber warnings, red hazards) with a colour-blind-safe palette for coverage states, keyboard navigation and ARIA labels. Offline base map from Natural Earth coastlines.

16 Demo script for the judges

#	Scene	What happens	What it proves
1	Chennai Port	A gillnet over rubble next to rock clusters is detected; a rock false alarm is rejected with its physics reason in the inspector.	Detection + explained noise filtering
2	Gulf of Mannar reef	A half-buried net without a shadow is found through its float chain.	Ghost-net specialisation
3	Kochi channel	Pipe and cylinder with heights in metres; an unlabelled object flagged as unknown man-made.	Physical measurement + open-set detection
4	Interference and dropouts	A dropout burst creates an insufficient area; a mirrored echo is vetoed.	Honesty about absence; artefact handling
5	Visakhapatnam post-cyclone	Before/after comparison lists new obstructions; a PDF work order is downloaded.	Disaster management value
6	SonarForge Lab	Judges change burial, range and roughness and watch PoD change.	Measured operating envelope

Closing the demo

End on the ablation table and the reliability diagram, then state clearly which numbers were measured on simulated, hybrid and real data. Credibility with NIOT scientists is the final differentiator.

17 Technology stack and hardware

Layer	Technology
Simulation and signal processing	Python 3.11, NumPy, SciPy, Numba, OpenCV, scikit-image
Machine learning	PyTorch, Ultralytics YOLO11n-seg, ResNet18 PatchCore-lite with FAISS, scikit-learn for calibration
Inference	ONNX Runtime (CPU); optional CUDA or TensorRT FP16/INT8 on Jetson Orin Nano
Formats and geo	pyxrf, custom JSF reader, h5py, pyproj, Shapely
Backend	FastAPI, WebSockets, SQLAlchemy with SQLite, ReportLab for PDF
Frontend	React, Vite, TypeScript, Tailwind CSS, MapLibre GL, Recharts, Zustand, SWR
Packaging	Docker Compose, Makefile, GitHub Actions CI

Minimum hardware: laptop with a 4-core CPU and 16 GB RAM. **Recommended edge:** NVIDIA Jetson Orin Nano on the survey vessel. **Training:** any single consumer GPU, or CPU in quick mode.

18 Build roadmap

Stage	Work	Acceptance check
M0	Repository, Makefile, Docker, CI	CI green
M1	Data contracts, XTF reader and writer, SNL, image logs	Round-trip tests pass
M2	SonarForge core	Gallery of every class and preset; shadow length within 5 % of formula
M3	Preprocessing chain	Slant-range correction within 1 px of truth
M4	CFAR and Net Signature Head	Precision-recall curves on simulated scenes
M5	Dataset builder and dataset card	Counts table generated
M6	YOLO training, ONNX export, anomaly bank	Metrics JSON saved
M7	Physics Verifier, calibration, confidence fusion	Reliability diagram
M8	Geotagging, dimensions, association, PoD, second look, change detection	CEP95 and PoD plots
M9	API, WebSocket streaming, review store, reports	API tests pass
M10	Dashboard, demo, final evaluation and ablation	Full demo runs offline

19 Risks and mitigations

Risk	Mitigation
Sim-to-real gap	Hybrid injection, domain randomisation, real fine-tuning, real held-out test, honest labelling of results by source.
Gated or unavailable datasets	Everything builds and demos on synthetic data alone; download scripts skip gracefully.
Flat nets with no shadow	Net Signature Head does not depend on shadows; the verifier is class-aware.
Poor navigation in real files	Pixel-only mode with explicit reasons; error radius grows with uncertainty.
Edge latency	CFAR pre-screen, small model, ONNX/TensorRT, tiling only where needed.
Over-confident outputs	Calibration, reliability diagrams, insufficient-evidence states, PoD labelled as simulation-estimated.
Licence issues	Licence file per dataset, AGPL note for Ultralytics, non-commercial data kept optional.

20 Impact, users and future scope

- **Environmental:** earlier removal of ghost nets that keep killing marine life and damaging reefs in the Gulf of Mannar, Palk Bay, Lakshadweep and Andaman waters.
- **Economic:** protection of fish stocks and fishing livelihoods; fewer propeller and intake fouling incidents in ports.
- **Disaster response:** faster, measured clearance of harbours and channels after cyclones, reopening ports sooner and more safely.
- **Scientific:** an open, physics-based SSS simulator and dataset for Indian seabeds that NIOT and academia can extend.
- **Strategic:** indigenous capability aligned with the Deep Ocean Mission and the Blue Economy vision.

Future scope

- Integration with NIOT AUVs and ROVs for autonomous second-look missions.
- Fusion with multibeam bathymetry for true 3D object heights.
- Synthetic Aperture Sonar support using the same pipeline.
- Gear-marking and fishing-activity data to predict net-loss hotspots.
- A national ghost-gear registry fed by confirmed detections and the data flywheel.

21 Final pitch

One-line pitch

SagarNetra is the eye of the ocean for India's ports and reefs: it manufactures the ghost-net sonar data the world does not have, finds nets even when they cast no shadow, proves every detection with acoustic physics, and tells responders exactly where to dive and which waters are truly clear.

Thirty-second version

Ghost nets and post-cyclone debris are invisible to cameras and slow for human sonar analysts. No ghost-net sonar dataset exists, so we built one by rendering physically correct nets into real seabed recordings. Our detector looks for the float chains and ropes that give nets away, a physics verifier explains every acceptance and rejection, and each target gets a calibrated confidence, a position with a 95 % error radius, and its length, width and height. Because we measured detection probability in simulation, our map also shows which areas are genuinely clear. It runs offline on the survey boat and compares surveys before and after disasters.

A Appendix: Antigravity build prompt

Copy everything in the dark box below into Google Antigravity as a single task. The prompt builds the complete SagarNetra prototype milestone by milestone, including the SonarForge simulator, the SagarNetra-SSS-IN v1 dataset, training, evaluation, the API, the dashboard and the demo. The same text is supplied as a separate file, antigravity-prompt.txt, for easy copying.

Before you run it

Set the environment variable HF_TOKEN if you have been granted access to gated Hugging Face datasets (Ghost Pot). Without it the prototype still builds and demos entirely on synthetic and hybrid data. Run `make train-quick` first so a working model always exists, then `make train` for the full run.

```
=====
ANTIGRAVITY BUILD PROMPT - "SagarNetra" (SIH 2026, Problem Statement 26057, NIOT / MoES)
AI-Powered Automated Underwater Marine Debris & Anomaly Detection using Side-Scan Sonar
=====

ROLE AND GOAL
You are a senior full-stack + machine-learning engineering agent working inside Google Antigravity.
Build a complete, runnable, offline-first prototype called "SagarNetra" (Sagar = ocean, Netra = eye).
SagarNetra ingests side-scan sonar (SSS) logs, detects ghost nets and other man-made seabed debris
(shipwrecks, pipes, cylinders, ropes, traps, unknown man-made objects), verifies every detection with
acoustic physics, geotags it with an honest error radius in metres, maps where the survey could and
could not see, and shows everything live on a dashboard with downloadable JSON / CSV / GeoJSON / KML /
PDF reports. It must also support a "Disaster Mode" (before/after cyclone survey comparison).

Work strictly milestone by milestone (M0 ... M10, section 18). After each milestone: run all tests,
fix failures, commit to git with a clear message, and append a short note to docs/PROGRESS.md.
Do not move to the next milestone while the current milestone's acceptance checks fail.

-----
0. NON-NEGOTIABLE RULES
-----
- 100 % offline at runtime. No cloud APIs, no hosted models, no telemetry. Internet is used ONLY by
  the explicit dataset download scripts in scripts/download/.
- Must run on a CPU-only laptop (Ubuntu 22.04 or Windows + WSL2, 16 GB RAM). CUDA GPU and NVIDIA
  Jetson Orin Nano (JetPack 6) are optional acceleration profiles selected by config.
- Never invent coordinates. If navigation is missing, position = null and a "reason" string is given.
- Never fabricate metrics. Every number shown in the UI or docs must be produced by a script in this
  repo and saved under artifacts/metrics/. Targets must be labelled "target", never "result".
- Tag every sample, label file and metric with source = "sim" | "hybrid" | "real".
- Respect dataset licences. Save each licence text in data/licenses/. Never commit raw data (gitignore).
  Gated Hugging Face datasets read the token from env var HF_TOKEN; if missing, skip with a clear message.
  THE WHOLE SYSTEM MUST BUILD, TRAIN (quick mode) AND DEMO USING SYNTHETIC DATA ALONE.
- Python 3.11, full type hints, pydantic v2 for all data contracts, ruff + mypy clean, pytest for every
  module. Deterministic generation: all seeds fixed and logged.
- Ultralytics is AGPL-3.0: record this in docs/MODEL_CARD.md.

-----
1. TECH STACK
-----
Backend / ML: numpy, scipy, numba, opencv-python-headless, scikit-image, scikit-learn, pyxtf, pyproj,
shapely, h5py, pandas, pydantic, fastapi, uvicorn[standard], websockets, sqlmodel (SQLite), reportlab,
torch + torchvision (CPU wheels default), ultralytics (YOLO11n-seg), onnx, onnxruntime
(onnxruntime-gpu / TensorRT optional), faiss-cpu, matplotlib, typer, rich, pyyaml.
Frontend: React 18 + Vite + TypeScript + Tailwind CSS + MapLibre GL JS + Recharts + Zustand + SWR.
Offline map: bundle Natural Earth 1:10m coastline + land polygons (public domain) as GeoJSON in
frontend/public/basemap/. OpenStreetMap raster tiles only as an optional "online" layer toggle.
Packaging: docker-compose.yml (backend + frontend) and a Makefile with targets:
setup, data-synth, data-real, dataset, train, train-quick, calibrate, export, evaluate, demo, test.

-----
2. REPOSITORY LAYOUT
-----
sagarnetra/
  backend/sagarnetra/
    io/          xtf_reader.py  xtf_writer.py  jsf_reader.py  image_log_reader.py  snl.py  schema.py
  preprocess/    quality.py    bottom_track.py  slant_range.py  gain.py    despeckle.py  motion.py  tiler.py
  detect/        cfar.py      net_signature.py  yolo_onnx.py   anomaly.py  fuse_candidates.py
  verify/        physics.py   rules.py
```

```

confidence/ quality_score.py calibration.py fusion.py
geo/ navigation.py layback.py georef.py error_budget.py dimensions.py
track/ association.py
coverage/ pod_table.py coverage_map.py second_look.py
change/ register.py change_detect.py
report/ json_report.py csv_report.py geojson_report.py kml_report.py pdf_workorder.py
api/ main.py routes_surveys.py routes_reports.py routes_lab.py ws.py db.py
pipeline.py (streaming orchestrator) cli.py (typer)
sonarforge/ seabed.py objects.py materials.py render.py noise.py motion.py nav.py
inject.py scenes.py export.py pod_sweep.py presets.yaml
training/ build_dataset.py train_yolo.py train_anomaly.py calibrate.py export_onnx.py
evaluate.py ablation.py
scripts/download/ ghostpot.py sctd.py sctd2.py kls2.py ai4shipwrecks.py s3simulator.py
noaa_usgs_sss.py seafloorai_subset.py marine_debris_fls.py
data/ raw/ interim/ datasets/ licenses/ demo_logs/ sources.yaml (gitignored except yaml+licenses)
frontend/ (Vite app)
artifacts/ models/ metrics/ plots/ pod/ reports/
docs/ ARCHITECTURE.md DATASET_CARD.md MODEL_CARD.md API.md DEMO_SCRIPT.md PROGRESS.md
tests/

```

3. DATA CONTRACTS (pydantic, backend/sagarnetra/io/schema.py)

Ping: ping_id, timestamp_utc, ship_lat, ship_lon, heading_deg, cog_deg, speed_mps, altitude_m, depth_m, heave_m, pitch_deg, roll_deg, yaw_deg, cable_out_m?, layback_m?, sound_speed_mps=1500, sample_intervals_s, frequency_hz, range_m, port: uint16[N], stbd: uint16[N], flags {dropout, nav_missing, altitude_jump, attitude_excess}.

Survey: survey_id, sensor_model, frequency_hz, range_m, crs, files[], start_utc, end_utc, line_ids[].

Detection: exactly the report schema in section 12.

Tile: tile_id, survey_id, side, ping_start, ping_end, ground_res_m, arrays (ch0..ch2), maps (ground_range_m, along_track_m, ping_index), quality_mask.

4. SONARFORGE - PHYSICS SIMULATOR + HYBRID INJECTOR (sonarforge/)

Goal: produce physically consistent side-scan waterfalls with perfect labels AND physical ground truth (height, size, burial, true lat/lon), because no public ghost-net SSS dataset exists.

4.1 Geometry. Per ping: towfish altitude $H = 2\text{--}20$ m and $8\text{--}20\%$ of range; range per side R in $\{30, 50, 75\}$ m; frequency 450 kHz or 900 kHz; $c = 1500$ m/s; slant sample spacing $dr = c \cdot dt / 2$; ping rate $\leq c / (2R)$; tow speed 1.5–2.5 m/s; horizontal beamwidth 0.4 deg (450k) / 0.2 deg (900k).

4.2 Seabed = 2.5D height map on 0.05 m grid, sum of:

- (a) low-frequency slope/undulation (fBm, $0\text{--}5$ deg), (b) sand ripples (wavelength $0.3\text{--}1.5$ m, amplitude $2\text{--}15$ cm, random orientation + phase noise), (c) rock clusters (Poisson-disc ellipsoids $0.3\text{--}3$ m wide, $0.2\text{--}2$ m high), (d) seagrass patches (low height, high-variance backscatter), (e) mud (smooth, low backscatter), plus a material map.

Presets (parameter DISTRIBUTIONS in presets.yaml, not fixed values):

EAST_COAST_SAND (Chennai / Visakhapatnam shelf style), WEST_COAST_MUD (Mumbai / Kutch style), REEF_RUBBLE (Gulf of Mannar / Lakshadweep style), SEAGRASS_BED (Palk Bay style), HARBOUR_FLOOR (port basin with tyres, blocks, chains, anchors).

4.3 Objects (each yields height map + material map + instance mask + physical metadata):

- ghost_net: gillnet or trammel fragment $2\text{--}60$ m. Headrope + footrope polylines with mesh panel between; floats every $0.5\text{--}2.0$ m (radius $4\text{--}10$ cm, air-filled or foam), sinkers every $0.3\text{--}1.0$ m; nylon twine weak backscatter with mesh-period texture (mesh $20\text{--}200$ mm); draped on seabed (height = seabed + $0\text{--}0.5$ m), burial $0\text{--}70\%$, optional bundled/balled mass (height up to 1.5 m), optional snag on a rock (raised section), random tears, optional entangled debris.
- rope_cable: polylines $5\text{--}100$ m, diameter $1\text{--}5$ cm, smooth curvature.
- pipe: cylinders diameter $0.2\text{--}1.2$ m, length $10\text{--}200$ m, straight or gently curved, burial $0\text{--}60\%$.
- cylinder: drums / barrels / mine-like shapes, diameter $0.3\text{--}1.0$ m, length $0.5\text{--}2.5$ m, any heading.
- wreck_debris: primitives for small boats $5\text{--}15$ m (upright or capsized), containers $6 / 12$ m, collapsed structure blocks, aircraft-like shapes; metal or wood materials.
- trap_pot: cages $0.5\text{--}1.2$ m (compatible with real crab-pot data).
- Confusers (trained as negative classes, NEVER reported as hazards): rock_cluster, sand_ripple, seagrass, fish_school (water column), biologic_mound.

4.4 Rendering per ping (numba). For each side cast rays across the 2.5D profile under the beam

(3 along-track sub-rays for beamwidth). For each slant bin:

```

I_dB = SL - 2*TL(r) + BS(theta_inc, material) + B(theta_dep)
TL(r) = 20*log10(r) + alpha*r      (alpha ~0.10 dB/m at 450 kHz, ~0.30 dB/m at 900 kHz)
BS = mu_material + 10*log10(cos^2(theta_inc))  (Lambert)
mu (dB): sand -27, mud -35, rock -17, metal -5, air float -3, foam float -12, nylon twine -30,
seagrass -25 (high variance), wood -20.

```

Occluded bins -> acoustic shadow (noise floor only). Then apply TVG, multiplicative speckle from a K-distribution (shape ν in $[1, 20]$; large $\nu \sim$ Rayleigh), additive noise floor, water column before first bottom return (with optional fish schools), nadir gap, surface return at $\sim 2 \cdot \text{depth}$, electrical interference stripes, rare port/starboard crosstalk ghosts (labelled), ping dropouts ($0\text{--}3\%$ single pings + occasional bursts of $5\text{--}40$ pings).

4.5 Motion applied to geometry BEFORE rendering: heave +/-0.5 m (0.1 Hz), pitch/roll +/-5 deg (0.05-0.2 Hz), yaw +/-3 deg, speed +/-20 %.

4.6 Navigation: lawn-mower survey lines 30-80 m apart, any heading, around real Indian demo sites: Chennai Port 13.10N 80.30E | Kochi 9.97N 76.24E | Visakhapatnam 17.69N 83.29E | Gulf of Mannar 9.10N 79.20E | JNPT Mumbai 18.95N 72.95E | Paradip 20.26N 86.68E. GPS noise sigma 1-3 m, layback from cable_out and depth, timestamps, attitude streams.

4.7 Scene outputs: raw waterfall (uint16 port/stbd), ping headers, instance masks (slant AND ground range), shadow masks, per-object truth (class, lat, lon, length, width, height, burial %, heading, material), seabed preset, SNR, roughness index. Save as SNL (HDF5) and optionally genuine XTF (5.2).

4.8 Hybrid injection (inject.py): take REAL seabed-only tiles (KLSG-II "seafloor", SCTD/SCTD2 backgrounds, NOAA/USGS side-scan, Ghost Pot negative regions). Estimate geometry (nadir line, altitude, range direction) from metadata or by detecting the first-return line; fit local backscatter + K-distribution speckle statistics; render the synthetic object's highlight and cast its shadow with the SAME geometry (away from nadir, length $L_s = h \cdot x_o / (H - h)$); attenuate the shadow region to the local noise floor; re-sample speckle to match the background; Poisson-blend edges. Output identical label format with source="hybrid".

4.9 PoD sweep (pod_sweep.py): reference targets REF_NET (5 m x 2 m gillnet fragment, 50 % buried, float spacing 1 m) and REF_CYL (0.6 m x 1.5 m cylinder). Sweep ground range (10 bins), H/R ratio (4 bins), roughness class (4), SNR (4); >= 50 random seeds per cell; run the FULL detection pipeline; save artifacts/pod/pod_table.npz (+ bin edges, pipeline version). Powers the clearance map.

5. INGESTION (backend/sagarnetra/io)

5.1 Readers: XTF via pyxtf (sonar channels, nav, attitude, altitude); minimal EdgeTech JSF reader (message type 80); SNL (our HDF5); image logs: PNG/TIFF/JPG waterfall + sidecar nav CSV (row_index or timestamp, lat, lon, heading, altitude, range_m, side_layout) or GeoTIFF mosaic (rasterio if installed). No navigation -> PIXEL-ONLY MODE: pixel detections, dimensions null, position null, explicit reason.

5.2 xtf_writer.py: minimal XTF writer (XTFFILEHEADER 1024 bytes incl. CHANINFO, XTFPINGHEADER 256 bytes, XTFPINGCHANHEADER 64 bytes, 16-bit samples) so SonarForge scenes become real XTF demo files. Round-trip test: write -> read with pyxtf -> samples and navigation identical.

5.3 Streaming: PingSource protocol; FileReplaySource with speed multiplier 1x-50x.

6. PREPROCESSING AND NOISE FILTERING (preprocess/)

- quality.py: per-ping flags: dropout (all-zero / constant / energy < 5th percentile), altitude jump (> 0.5 m ping-to-ping), nav gap (> 2 s), attitude rate excess. Interpolate single-ping dropouts; mark bursts NO_DATA (never hallucinate content).
- bottom_track.py: first-return detection per side (smoothed energy gradient) + Kalman-smoothed altitude; fall back to header altitude; report disagreement.
- slant_range.py: $x = \sqrt{r^2 - H^2}$; remove water column; resample to a uniform ground grid (default 0.10 m/px) -> this normalises varying pixel resolution across sonars and ranges.
- motion.py: along-track resampling by distance travelled (speed over ground) to 0.10 m; heading/yaw compensation; per-ping H handles heave.
- gain.py: empirical gain normalisation (mean per ground-range bin over sliding 200-ping window), log-compress, clip to [0, 1].
- despeckle.py: SRAD or enhanced Lee filter (7x7, numba). Keep the raw layer as well.
- tiler.py: 512 x 512 tiles, 25 % overlap, per side, carrying ground_range, along_track, ping_index.
- Feature stack fed to the detector INSTEAD of RGB:
 - ch0 = normalised despeckled intensity, ch1 = CFAR shadow probability, ch2 = ridge/line map (Sato filter, scales 1-3 px). Explain the choice in MODEL_CARD.md.

7. DETECTION (detect/)

- cfar.py (Stage A): OS-CFAR (guard 3, train 16, Pfa $1e-3$) for highlights and inverse CFAR for shadows; pair highlight blobs with a shadow on the FAR-range side within 0.2-15 m -> candidates.
- net_signature.py (Net Signature Head):
 - (a) LoG blob detection of small bright points 0.08-0.5 m (floats / sinkers);
 - (b) chain finder: graph with edges 0.3-3 m -> MST -> longest smooth paths; score spacing regularity (coefficient of variation < 0.35 = good) and curvature smoothness;
 - (c) ridge skeleton length >= 2 m (ropes, head/foot lines);
 - (d) mesh texture energy (Gabor bank tuned to 2-8 px periods).
 Output S_net in [0,1] + heatmap + human-readable evidence ("chain of 14 points, spacing 0.9 m").
- yolo_onnx.py (Stage B): YOLO11n-seg trained on the 3-channel feature stack, imgsz 512, exported to ONNX (opset 17), run with onnxruntime; optional FP16/INT8 TensorRT on Jetson. Hazard classes: ghost_net, rope_cable, pipe, cylinder, wreck_debris, trap_pot. Confuser classes (trained, suppressed from reports): rock_cluster, sand_ripple, seagrass, fish_school.
- anomaly.py: PatchCore-lite: frozen ResNet18 (layer2+layer3) patch features from seabed-only tiles, 1 % coreset, faiss KNN -> anomaly map. High anomaly + no hazard class + no confuser class -> class "unknown_manmade" (important for post-disaster debris no class list covers).
- fuse_candidates.py: union of YOLO detections, CFAR pairs with high anomaly, and Net-Signature hits (S_net > 0.6); NMS by mask IoU 0.5.

8. PHYSICS VERIFIER (verify/)

Per detection compute highlight extent, ground shadow length L_s , object ground range x_o , altitude H .
Height $h = H * L_s / (x_o + L_s)$. Rules return +1 consistent / 0 not assessable / -1 inconsistent + reason:
R1 class-height prior (m): net 0-1.5, rope 0-0.3, pipe 0.1-1.5, cylinder 0.2-2.0, wreck 0.5-15, trap 0.3-1.2
R2 highlight-before-shadow order (shadow must lie at greater range than highlight)
R3 port/starboard mirror (same ping +/-3, same range +/-0.3 m on opposite side -> interference)
R4 along-track persistence consistent with the expected beam footprint (single-ping spikes = noise)
R5 water-column target (range < first bottom return) -> fish_school / suspended, not seabed debris
R6 nadir band or ~2x depth surface-multipath band -> artefact
R7 resolution adequacy: object length / along-track footprint ($x * \theta_{h_}$) ≥ 3 , else insufficient
R8 local slope > 5 deg -> height unreliable, widen position error
Nets WITHOUT a shadow are NOT penalised by R1/R2 (class-aware; flat nets cast almost no shadow).
Output V_{phys} in [-1, 1] and the full rule log (shown in the UI and the PDF).

9. CONFIDENCE (confidence/)

- calibration.py: temperature scaling of class logits on the calibration split; ECE (15 bins) and reliability diagram before/after.
- quality_score.py: Q_{obs} in [0,1] = product of: dropout-free fraction in object window; altitude factor (best at H/R 0.08-0.20); range-band factor (penalise nadir zone and outer 10 %); attitude stability factor; resolution-adequacy factor (R7).
- fusion.py: Hazard Confidence (0-100 %)
 $C = 100 * \text{sigmoid}(w_0 + w_1 * \logit(p_{cal}) + w_2 * Q_{obs} + w_3 * V_{phys} + w_4 * S_{net} + w_5 * \min(\text{views}, 3))$
weights fitted by logistic regression on the hybrid+real validation split and checked with a reliability diagram. ALWAYS also return the components: p_{cal} , Q_{obs} , V_{phys} , S_{net} , $position_certainty (= 1 / (1 + r_{95_m} / 10))$.
- Status: CANDIDATE ($C \geq 60$) | LOW_EVIDENCE ($35 \leq C < 60$) | REJECTED ($C < 35$, kept in the log) | INSUFFICIENT_EVIDENCE (cell level, from coverage).

10. GEOTAGGING AND DIMENSIONS (geo/, track/)

- WGS84 <-> UTM with pyproj; auto-select EPSG:32643 / 32644 / 32645 for Indian longitudes.
- Towfish: use layback if given, else $L = \sqrt{\max(\text{cable_out}^2 - \text{depth}^2, 0)}$; towfish = GPS antenna - vessel offsets - L along course (use COG when speed > 0.5 m/s, else heading).
- Target: $E = E_f + s * x * \cos(\psi)$, $N = N_f - s * x * \sin(\psi)$ (ψ clockwise from north; $s = +1$ starboard, -1 port); along-track position by timestamp interpolation of the object's centre ping.
- Error budget: $\sigma_{\text{map}}^2 = \sigma_{\text{gps}}^2 + \sigma_{\text{L}}^2 + (x * \sigma_{\psi})^2 + \sigma_x^2 + (v * \sigma_t)^2$
defaults: σ_{gps} 2 m, σ_{L} $0.1 * L + 1$ m, σ_{ψ} 1 deg, σ_x 0.5 px + slope term, σ_t 0.1 s. Report $r_{95_m} = 2.45 * \sigma_{\text{map}}$ and a method string.
- Dimensions: oriented minimum-area rectangle of the ground-range mask -> length, width, orientation (deg from north), area m^2 ; height from the verifier (or null + reason); outline polygon in WGS84.
- association.py: merge detections from different lines when Mahalanobis distance < 3 (using both error ellipses) and classes are compatible; keep per-view evidence; views are correlated, so the fusion caps their weight.

11. COVERAGE, PoD, SECOND LOOK AND DISASTER MODE

- coverage_map.py: 1 m grid; each cell stores the best PoD (MAX over passes - conservative because passes are correlated) for REF_NET and REF_CYL by interpolating pod_table with the cell's ground range, H/R, local roughness (texture variance) and SNR. States: NOT_SURVEYED, INSUFFICIENT (PoD < 0.5 or quality flags), SURVEYED_CLEAR (PoD ≥ 0.8 and no candidate), CANDIDATE. Report km^2 per state. Label PoD as "simulation-estimated".
- second_look.py: for LOW_EVIDENCE targets and INSUFFICIENT clusters propose a line with a perpendicular heading that places the target at 30-60 % of range; export GeoJSON + GPX.
- change/: register before/after mosaics (georeference + ECC refinement on overlap), change map = normalised log-ratio + PatchCore feature distance, components > 1 m^2 -> NEW_OBSTRUCTION or REMOVED; each goes through verifier + geotag + report. Ship 3 synthetic pre/post-cyclone pairs:
(1) Chennai Port basin: sunken fishing boat + container + net mass,
(2) Kochi channel: collapsed pier blocks + pipe displacement,
(3) Visakhapatnam fishing harbour: several capsized boats + ropes.

12. REPORTS (report/)

JSON (schema_version "1.0"):

```
{ report_id, generated_utc, software_version, model_version,
  survey { survey_id, sensor_model, frequency_hz, range_m, files[], crs },
  coverage { surveyed_km2, clear_km2, insufficient_km2, not_surveyed_km2 },
  detections: [ {
    id, class, subtype, status, hazard_confidence,
    components { p_cal, q_obs, v_phys, s_net, position_certainty },
    rule_log [ { rule, verdict, reason } ],
    position { lat, lon, crs: "EPSG:4326", r95_m, method } OR null + position_reason,
```

```

    dimensions_m { length, width, height|null, area_m2, orientation_deg },
    bbox_px { side, ping_start, ping_end, sample_start, sample_end },
    polygon_wgs84, views, first_seen_utc, source_file, priority, recommended_action,
    review { status: unreviewed|confirmed|rejected|unsure, reviewer, note } } ] }
CSV: one row per detection, flattened fields. GeoJSON: FeatureCollection with Point (+ r95_m),
Polygon outlines, coverage polygons and second-look LineStrings. KML optional.
PDF Work Order (reportlab): cover, map snapshot, ranked target table, one evidence page per target
(crop, mask, shadow, height diagram, rule log, search box = square of side 2*r95), second-look plan.
Priority = (C/100) * severity(class; x1.5 if within 200 m of a user-supplied channel or reef polygon)
          * 1/(1 + r95_m/10).

```

13. API (FastAPI)

```

POST /api/surveys                multipart upload (XTF/JSF/SNL or image + nav CSV) -> survey_id
POST /api/surveys/{id}/run?replay_speed=10
WS   /ws/surveys/{id}            streams {type: ping_batch|tile_done|detection|coverage_update|progress}
GET  /api/surveys/{id}/detections
PATCH /api/detections/{id}/review
GET  /api/surveys/{id}/report.{json|csv|geojson|kml|pdf}
GET  /api/coverage/{id}
POST /api/change                 {before_id, after_id}
POST /api/sonarforge/scene       params -> preview PNG + predicted PoD
GET  /api/health                 device (cpu/cuda/tensorrt), model + pipeline versions

```

14. DASHBOARD (frontend/)

Theme: dark ops console. Colours: navy #0B2545 background, teal #13A89E accent, amber #F2A900 warnings, red #D64545 hazards, white/grey text. Fonts: Inter + JetBrains Mono (bundled locally).

Screens:

- (1) Upload & Mission Control: drag-and-drop upload; dual waterfall canvas (port | starboard) scrolling live with coloured masks; MapLibre map with ship track, towfish track, detection pins with r95 circles, coverage layer (grey not surveyed, amber insufficient, green clear); counters (pings, km², detections per class, processing pings/s, device).
- (2) Target Inspector drawer: crop with mask + shadow overlay, height diagram, component bars, rule log ("why"), buttons Confirm / Reject / Change class / Unsure (stored for active learning).
- (3) Coverage & Clearance: PoD heatmap (REF_NET / REF_CYL), second-look lines, clearance summary.
- (4) Disaster Compare: before/after swipe slider, NEW_OBSTRUCTION list with zoom-to.
- (5) SonarForge Lab: sliders (seabed preset, net length, burial, range, altitude, SNR) -> generated waterfall + live PoD curve.
- (6) Reports: download JSON / CSV / GeoJSON / KML / PDF, JSON preview.

Accessibility: keyboard navigation, ARIA labels, colour-blind-safe palette for states. Primary layout >= 1280 px, still usable at 1024 px.

15. DATASET "SagarNetra-SSS-IN v1" AND TRAINING (training/)

build_dataset.py composes:

```

S 20,000 simulated tiles (all 5 seabed presets, all classes, 450/900 kHz, all ranges)
H 15,000 hybrid tiles (synthetic objects injected into REAL seabed backgrounds)
R real public labelled tiles re-mapped to our taxonomy:
  Ghost Pot crab pots -> trap_pot; SCTD / KLSG-II ship -> wreck_debris(subtype ship);
  aircraft -> wreck_debris(subtype aircraft); SCTD2 traps -> trap_pot;
  AI4Shipwrecks masks -> wreck_debris(subtype ship)
N 10,000 real seabed-only tiles (negatives, anomaly bank, false-positive suppression)
D 300 synthetic before/after disaster pairs
X 6 synthetic XTF demo logs (one per demo site) with full navigation and attitude
Formats: YOLO-seg txt + COCO JSON + per-tile metadata JSON (source, preset, sensor, frequency,
ground_res_m, altitude, SNR, roughness, per-object physical truth).
Splits: by scene seed / survey / site / physical object - NEVER random crops of the same image.
Keep TEST_REAL_HELDOUT (entire unseen sites) untouched until final evaluation.
docs/DATASET_CARD.md: counts per class x source x split, licence per part, known biases.

```

Augmentations ALLOWED: port<->starboard mirror across nadir (shadows still point away from nadir), along-track flip, along-track stretch 0.8-1.2, gain/TVG jitter, gamma, K-distribution speckle re-sampling, dropout stripes, range-dependent blur, rotation +/-5 deg.

Augmentations FORBIDDEN (break sonar physics): 90/180 deg rotations, flips that make shadows point toward nadir, colour jitter, mosaic that mixes sides or ranges.

Curriculum: 60 epochs on S+H, then 40 epochs fine-tune on S+H+R (R oversampled x3); imgsz 512; batch 16 (CPU: batch 4); AdamW lr 1e-3 cosine; early stop on val mask mAP50-95.

train-quick: 3,000 tiles, few epochs, CPU-friendly, so the prototype is ALWAYS demoable.

train_anomaly.py: PatchCore bank from N-train. calibrate.py: temperature + fusion weights.

export_onnx.py: ONNX vs PyTorch max abs diff < 1e-3; INT8 re-check of ghost_net recall.

16. EVALUATION (training/evaluate.py, ablation.py) -> artifacts/metrics/*.json + plots

mAP50 and mAP50-95 (box + mask) per class and per source; ghost_net recall at 1 / 2 / 5 false

alarms per km²; ECE before/after calibration; geolocation CEP50 / CEP95 against synthetic truth; height MAE (and % error for h > 0.3 m); PoD curves vs range / burial / roughness; latency per tile and pings/s on the current device.
Ablation table: YOLO(GB-like single channel) -> +feature stack -> +Net Signature -> +Physics Verifier -> +Anomaly head -> +confidence fusion.
TARGETS (show as targets): ghost_net recall >= 0.80 at <= 2 FA/km² on hybrid test; ECE <= 0.05; CEP95 <= 5 m with standard GPS on synthetic truth; <= 150 ms per 512 px tile on laptop CPU.

17. DEMO (make demo + docs/DEMO_SCRIPT.md)

make demo: generate (if missing) the 6 XTF demo logs and 3 disaster pairs, load the quick model, start backend + frontend, open the dashboard. Scripted scenes:

- (a) Chennai Port: gillnet draped over rubble beside rock clusters -> rock false alarm rejected with a physics reason shown in the Inspector.
- (b) Gulf of Mannar reef: half-buried net found by its float chain (Net Signature evidence).
- (c) Kochi: pipe + cylinder + an "unknown_manmade" object flagged by the anomaly head.
- (d) Dropout burst + port/starboard interference -> INSUFFICIENT area + interference veto.
- (e) Visakhapatnam post-cyclone before/after -> NEW_OBSTRUCTION list + PDF work order.
- (f) SonarForge Lab: change burial / range / roughness and watch the PoD curve update live.

18. BUILD ORDER - MILESTONES WITH ACCEPTANCE CHECKS

- M0 Repo scaffold, Makefile, docker-compose, CI (ruff, mypy, pytest) -> CI green
 - M1 Data contracts, XTF reader + writer, SNL, image-log reader -> round-trip tests pass
 - M2 SonarForge core (seabed, objects, render, noise, motion, nav) -> gallery PNG of every class x preset; tests: shadow direction correct, shadow length vs formula within 5 %
 - M3 Preprocessing chain -> slant-range correction within 1 px of synthetic truth; before/after plots
 - M4 CFAR + Net Signature Head -> PR curves on sim scenes
 - M5 Dataset builder (S, H, and R/N when downloaded) + DATASET_CARD.md -> counts table generated
 - M6 YOLO training (quick + full), ONNX export, anomaly bank -> metrics JSON saved
 - M7 Physics verifier, calibration, confidence fusion -> reliability diagram
 - M8 Geotagging, dimensions, association, coverage/PoD, second look, change -> CEP95 + PoD plots
 - M9 FastAPI + WebSocket streaming + SQLite review store + all report formats -> API tests pass
 - M10 React dashboard (6 screens) + make demo + DEMO_SCRIPT.md + final evaluation + ablation
- At the end print: what works, measured metrics (from artifacts/metrics), and known limitations.

19. REAL DATA DOWNLOAD SCRIPTS (scripts/download/, sources in data/sources.yaml)

Each script prints source URL, licence, expected size; verifies checksums where available; writes into data/raw/<name>/ and the licence into data/licenses/<name>.txt.

- Ghost Pot Side-Scan Sonar Detection Dataset (Hugging Face, PING ecosystem / Univ. of Delaware collaborators): ~6,674 images, Humminbird SSS, crab pots, JSONL boxes, GPL, gated -> HF_TOKEN.
- SCTD 1.0 (GitHub, Pascal VOC XML) and SCTD2 (GitHub, Klein SSS, small traps, 320x320, segmentation; README says detection needs re-annotation - derive boxes from masks).
- SeabedObjects-KLSG-II (GitHub): 66 airplane, 487 ship, 578 seafloor images.
- AI4Shipwrecks (University of Michigan): 286 SSS images from 28 wreck sites, pixel masks.
- S3Simulator (synthetic SSS, 7,721 ship/plane images): pre-training + comparison only.
- NOAA NCEI hydrographic side-scan and USGS coastal & marine side-scan data releases (raw XTF where available): backgrounds, ingestion and navigation/geotag tests. Take survey IDs/URLs from data/sources.yaml (user-editable), do not hard-code.
- SeafloorAI (Hugging Face, CC BY-NC-SA 4.0, request-based): optional small subset for seabed texture statistics only; mark non-commercial.
- Marine Debris FLS datasets (Zenodo): optional, kept separate, NEVER mixed into SSS training.

If a URL has changed, search for the dataset by its exact name, confirm the licence, update data/sources.yaml and continue. If a dataset is unavailable, log it and continue with the rest.

DELIVERABLES: runnable repo, trained quick model + ONNX, 6 XTF demo logs, 3 disaster pairs, dashboard, sample reports in artifacts/reports/, docs (ARCHITECTURE, DATASET_CARD, MODEL_CARD, API, DEMO_SCRIPT, PROGRESS), and measured metrics + ablation in artifacts/metrics/.

B Appendix: References and data sources

1. Smart India Hackathon 2026, Problem Statement 26057: AI-powered automated underwater marine debris and anomaly detection using side-scan sonar. National Institute of Ocean Technology (NIOT), Ministry of Earth Sciences.
2. Report A reviewed: "Marine Debris Intelligence: Research and Product Opportunity Report".
3. Report B reviewed: "Detecting Ghost Nets and Marine Debris in Side-Scan Sonar Images: SonarTwin Solution Report".
4. Ghost Pot Side-Scan Sonar Detection Dataset, Hugging Face (gated).
5. SCTD and SCTD2 side-scan sonar target datasets, GitHub.
6. SeabedObjects-KLSG-II side-scan sonar dataset, GitHub.
7. AI4Shipwrecks: side-scan sonar shipwreck segmentation dataset, University of Michigan.
8. S3Simulator: synthetic side-scan sonar dataset for ship and plane targets.
9. NOAA National Centers for Environmental Information, hydrographic survey side-scan data; USGS Coastal and Marine Hazards and Resources data releases.
10. SeafloorAI: large-scale seafloor mapping dataset, Hugging Face (CC BY-NC-SA 4.0).
11. Marine Debris Forward-Looking Sonar datasets, Zenodo.
12. Triton Imaging, eXtended Triton Format (XTF) specification; EdgeTech JSF data format.
13. Blondel, P. The Handbook of Sidescan Sonar. Springer.
14. Lurton, X. An Introduction to Underwater Acoustics: Principles and Applications. Springer.
15. Guo, C. et al. On Calibration of Modern Neural Networks (temperature scaling). ICML 2017.
16. Roth, K. et al. Towards Total Recall in Industrial Anomaly Detection (PatchCore). CVPR 2022.
17. Ultralytics YOLO11 documentation (AGPL-3.0).
18. FAO / GGGI guidance on abandoned, lost or otherwise discarded fishing gear (ALDFG).

SagarNetra solution report for SIH 2026 PS 26057. All performance figures are design targets unless produced by the prototype's own evaluation scripts.